

current electronic technology in that it uses the concept of blocks of computation units, such as adders, stores and wires. The difference is that the building blocks are not transistors, but Quantum Cells that leverage quantum effects. The other kind makes use of quantum physics to change computing, the way electronic computers surpassed the old mechanical variety. Of the two, QCA is simpler and more likely to be in use commercially in the medium term.

Quantum Cellular Automata

Quantum Cellular Automata comprises a series of tiny Quantum cells arranged in patterns to harness their interactions for computing. Each cell is composed of a group of four or five quantum dots. A quantum dot is a nano-structure, created to trap and hold a single electron. It consists of a highly insulating material, enclosing a very small island of conducting material. This conducting material can hold an electrical charge—an electron. As temperatures increase, the size of the quantum dot must decrease to be stable. Current research indicates that these quantum dots would have to be 1 nanometer (nm) in size at room temperatures. Compare this to current CPU fabrication at 130 nm or 0.13 microns.

The quantum dots are arranged as shown in Figure 2, to create one Quantum cell. There are only two electrons in one cell, though you see four quantum dots ready to hold them. The quantum dots are divided into two pairs, each sharing an electron between them via a quantum-tunnelling junction. The electron can take advantage of quantum tunnelling, to instantly appear in the paired quantum dot, without ever needing to overcome the resistance of the insulation. That's quantum weirdness for you.

Both electrons have a negative charge, and repel each other so when one electron takes up residence at a dot, the other must go to the diagonally opposite corner. This way you have a simple binary state, where the electrons can be made to toggle between the two diagonals,



Figure 1: Quantum dots can take various physical shapes.

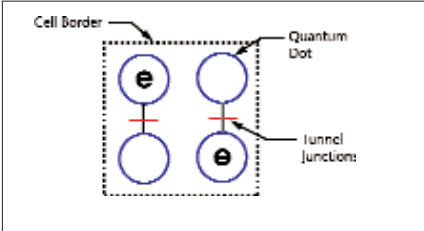


Figure 2: Quantum dots placed near each other form Quantum cells

forming one bit of storage.

Computing with QCA

When you have a row of bits lined up in your computer's RAM, they normally do not interact with each other at all. Each stores its state on its own, and minds its own business. But quantum cells are different. Very interesting effects appear when you bring two cells together. If you line up two cells, edge to edge, they will both assume the same diagonal. Again this is because electrons prefer not to stay close to each other. The charge one—the electron in the first cell—repels the electron in the second cell, forcing it into the opposite quantum dot.

If you lay down a line hundred cells

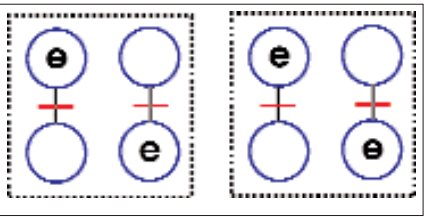


Figure 3: Quantum cells in proximity affect each other.

long, and flip the electrons at one end, all the cells along the line will also flip to that state. You would have created a simple wire that transfers the bit set at one end to the other end. This property of quantum cells can be used to create logic circuits for Boolean logic.

While quantum cells have yet to be fabricated at anywhere near room temperatures, theoretical researchers have been busy working out the mathematics and logic needed to use QCA. Already QCA have been shown to implement, exactly the same logic as transistor-based circuits do now.

This begs the question that if QCA are just another way to implement what we have today, why should we be bothered? One direct trend in transistor-based computing is the need to stuff more and more transistors close together, to boost speeds

and efficiency. Transistors are already so small that quantum effects are beginning to show themselves. If smaller devices are to be made, they must harness quantum effects, and not just work around them.

Current leakage in conventional circuits not only effects the efficiency of chips, it also limits how small and how near the components of the circuit can be before they begin interfering with each other. QCA do not suffer from this problem because they depend on the fact that there is zero leakage, only manipulation of trapped electrons. Both these advantages mean that Moore's law can continue to progress into sub-atomic realms.

Quantum bits and superposition

In conventional computing, we are accustomed to the term bit representing a state, usually denoted as 0 or 1. Any physical object capable of holding two steady states can be used to hold a bit. For example, bits are stored as the state of a magnetic domain on your hard disk surface, or as a capacitance charge in RAM, or as we saw in QCA, the position of electrons in a Quantum Cell.

As we reduce the size of physical storage for one bit, approaching sub-atomic particle sizes, a strange phenomenon takes over—the particle begins to exhibit Quantum Mechanical behaviour. The most odd manifestation of this is that the bit can suddenly be 0, 1 or both at the same time! To be precise, the state of the bit is a probability of being 0 or 1. As long as the bit is isolated from the rest of the world, its state is a probability distribution. As soon as the real world peeps in, it drops into one determinable state. We cannot accurately predict which state it will be in, without actually observing it. This behaviour is called superposition, and is one of the big mysteries of quantum theory.

If your remember Young's experiment

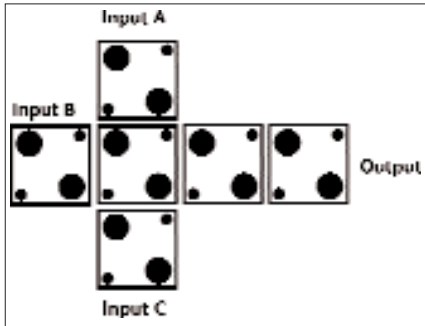


Figure 4: Logic gates made from QCA perform the same operations as conventional transistors